

Coproduct.

Def. Let $\{X_i | i \in I\}$ be a collection of Topological Spaces.

Consider the set $\coprod_{i \in I} X_i \stackrel{\text{def}}{=} \bigcup_{i \in I} X_i \times \{i\}$

and for each $i \in I$, Consider the injective map

$$\mathcal{Q}_i: X_i \rightarrow \coprod_{i \in I} X_i: x \mapsto (x, i).$$

The Topology $\mathcal{Q}$ on $\coprod_{i \in I} X_i$ which is the largest Topology such that

for all $i \in I$, $\mathcal{Q}_i$ is Continuous is called the disjoint union topology. Simply, denote $\coprod X_i$.

Prop. Let $U \subset \coprod X_i$ be a subset of the disjoint union space. Then,

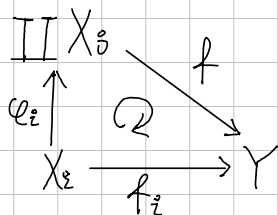
U is open set of $\coprod X_i$ if and only if for all $i \in I$, $\mathcal{Q}_i^{-1}[U] \subset X_i$ is open.

Proof. Since the Topology on $\coprod X_i$ is the final Topology by $\{\mathcal{Q}_i | i \in I\}$, clear. $\square$

Diagrammatically, for given a space Y and Continuous maps $f_i: X_i \rightarrow Y$ ($i \in I$), there exists a unique Continuous map

$$f: \coprod X_i \rightarrow Y: (x, i) \mapsto f_i(x)$$

which satisfies that for all $i \in I$, $f_i = f \circ \mathcal{Q}_i$.



Adjunction space.

Def. Let X and Y be Topological spaces, and $A \subset X$ be subspace. For a Continuous map $f: A \rightarrow Y$, define an equivalent relation on $X \amalg Y$ which is generated by for all $a \in A$, $a \sim f(a)$.

Then, the quotient topology

$X \cup_f Y \stackrel{\text{def}}{=} (X \amalg Y) / \sim$ is called the adjunction space by f .

And, the map f is called the attaching map.

Note: Specifically, the equivalent relation $\sim$ on $X \amalg Y$ is consisted by:

$$[z] = \begin{cases} f^{-1}[f[z]] \cup f[z] & \text{if } z \in A \\ \{z\} & \text{if } z \in X \setminus A \\ f^{-1}[f[z]] \cup \{z\} & \text{if } z \in f[A] \\ \{z\} & \text{if } z \in Y \setminus f[A] \end{cases}$$

In language of function, define the Canonical embeddings

$$\begin{aligned} \iota_X: X &\rightarrow X \cup_f Y & \text{and} & & \iota_Y: Y &\rightarrow X \cup_f Y \\ x &\mapsto (x, 0) & & & y &\mapsto (y, 1) \end{aligned}$$

$$\begin{array}{ccccc} X & \xhookrightarrow{\iota_X} & X \amalg Y & \xhookleftarrow{\iota_Y} & Y \\ & \searrow & \downarrow \pi & \swarrow & \\ & & X \cup_f Y & & \end{array}$$

Lemma. Let $A \subset X$ be a closed (or open) set, and $f: A \rightarrow Y$ be continuous. Then, the quotient map $\pi: X \amalg Y \rightarrow X \cup_f Y$ induces a Homeomorphism

$$\pi|_Y: Y \xrightarrow{\cong} \pi[Y] \quad (\text{Canonical Copy})$$

Moreover, $\pi[Y] \subseteq X \cup_f Y$ is a Closed (or open) set.

$$\begin{array}{ccccc} X & \xhookrightarrow{\quad \iota_X \quad} & X \amalg Y & \xhookleftarrow{\quad \iota_Y \quad} & Y \\ & \searrow & \downarrow \pi & \swarrow & \\ & & X \cup_f Y & & \end{array}$$

Examples.

Mapping Cylinder.

Let $f: X \rightarrow Y$ be a Continuous map. Then, for an interval $I = [0, 1]$,
let $\tilde{f}: X \times \{1\} \rightarrow Y: (x, 1) \mapsto f(x)$. Define the adjunction space

$$M_f \stackrel{\text{def}}{=} (X \times I) \cup_{\tilde{f}} Y$$